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Requirements Analysis Report: AI-Based Car Diagnostic Mobile Application

Prepared by: Business Development Department **Project:** CEF440 - Internet Programming (Group 11)

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1. Introduction

This report provides a strategic analysis of the requirements gathered for an AI-powered mobile diagnostic tool. Our research indicates a significant knowledge gap, with 62% of car owners unable to interpret critical dashboard warnings. This application aims to bridge the gap between professional mechanical expertise and car owner's practical challenges by simulating a mechanic's "eyes and ears" through a smartphone. The goal is to foster a proactive maintenance culture while respecting the safety boundaries of professional repair.

2. Classification of Requirements

A. Functional Requirement

- **Dashboard Light Recognition:** Use of the phone camera to scan, identify, and explain warning symbols.
- **Engine Sound Diagnosis:** Recording and analyzing engine noise via the microphone to identify faults like knocking or squealing.
- **Fault Explanation & Urgency:** Providing clear descriptions of issues with associated urgency levels (Immediate, Moderate, Minor).
- **Multimedia Guidance:** Integration of video tutorials and step-by-step repair guides for basic tasks.
- **History & Records:** A module to save past scans and diagnoses for future reference by the user or a mechanic.

B. Non-Functional Requirements

- **Cross-Platform Compatibility:** The application must function on both Android and iOS devices.
- **Offline Accessibility:** Core diagnostic features must remain functional without an active internet connection to accommodate local infrastructure constraints.
- **High Accuracy & Reliability:** The system must meet high accuracy expectations to overcome the "Accuracy-Trust Paradox".
- **Bilingual Interface:** Support for both English and French to cater to the diverse Cameroonian demographic.
- **Simple UX/UI:** The interface must be jargon-free and intuitive for non-technical users.

3. Review and Analysis

- **Completeness:** The requirements cover the core diagnostic journey from fault detection to repair guidance. However, the "Safety and Liability" section suggests a need for more robust legal disclaimers.
- **Clarity:** The sound-fault mapping (e.g., Hissing → leaks) provides excellent clarity for the AI development team. The distinction between "basic maintenance" and "critical failures" is well-defined to manage user expectations.
- **Technical Feasibility:** While camera-based recognition is standard, high-accuracy sound analysis in noisy environments (busy Cameroonian streets) poses a significant technical challenge. Offline AI processing will also require optimized machine learning models to run on mid-range devices.
- **Dependency Relationships:**
  - The **Sound Diagnosis** feature depends on the **Mechanic-Validated Dataset** for accurate mapping.
  - The **Offline Functionality** depends on **Preloaded Local Databases**.
  - The **Video Tutorials** feature is dependent on **Cloud/Internet Service Providers** for streaming, unless cached locally.

4. Inconsistencies, Ambiguities, and Missing Information

- **Inconsistencies:** There is a conflict between user desire for "complex fixes" and mechanic advice to restrict guidance to "basic maintenance". We have strategically decided to follow the mechanic's advice for safety.
- **Ambiguities:** The term "Moderate Trust" among users is ambiguous. It is unclear what specific accuracy threshold (e.g., 90% vs 99%) will satisfy the user base enough to rely on the app during a crisis.
- **Missing Information:**
  - **Data Privacy:** There is no mention of how user vehicle data or recorded audio will be stored or protected.
  - **Monetization:** The document does not specify if the app will be free, subscription-based, or ad-supported.
  - **Hardware Minimums:** Specific smartphone requirements (camera resolution or microphone sensitivity) are not yet established.

5. Requirement Prioritization Matrix

We have prioritized features based on their impact on user safety and technical ease of implementation.

Priority	Requirement	Reason
High	Dashboard Light Recognition	High user confusion (62%) and high feasibility via image recognition.
High	Offline Functionality	Critical for the Cameroonian market with inconsistent connectivity.
Medium	Sound-Fault Mapping	Highly desired but technically difficult in noisy environments.
Medium	Basic Video Tutorials	Necessary for user empowerment but requires high bandwidth for initial download.
Low	Scan History	Valuable for long-term use but not critical for immediate diagnostic needs.

6. Conclusion

The requirement gathering phase has established a robust foundation for a niche product tailored to the Cameroonian automotive market. By prioritizing reliability and transparency, the application will serve as a vital intermediary between car owners and professional mechanics. Our strategic pivot toward a "Hybrid Assistance" model ensures that we provide immediate value through AI while mitigating risks by referring critical failures to professionals. Moving forward, we must address the missing data privacy and hardware specifications to ensure a successful market launch.